Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

1. COURSE: DATA ANALYTICS AND VISUALIZATION - 24DSB3101

A.Y. 2026 – 2027

TITLE OF THE PROJECT	Intelligent Customer Churn Prediction Using Data Analytics and Classification Models	
S. No.	**University ID**	**Name**
1	2420030540	Allipuram Akhil
2	2420030440	Devineni Surya
Name of the Guide		DR.K.Swapnika

Abstract

Abstract

Customer churn is a significant challenge for organizations, as the loss of existing customers can affect revenue, customer relationships, and long-term business performance. Predicting customers who are likely to churn can help organizations identify potential risks and take appropriate retention measures. However, customer churn datasets contain a combination of demographic, service, usage, contract, and billing attributes that may involve complex relationships and nonlinear patterns. To address this challenge, this project proposes an intelligent customer churn prediction framework based on deep learning, exploratory data analysis, feature engineering, explainable artificial intelligence, and risk-level classification.

The proposed framework begins with data preprocessing, including missing-value handling, categorical feature encoding, feature scaling, and feature engineering. Additional features such as account age, average monthly spending, and charges per tenure are derived to improve the representation of customer behavior. Exploratory Data Analysis (EDA) is performed to examine churn distribution, customer characteristics, feature relationships, and important patterns in the dataset. The processed dataset is divided into training, validation, and testing sets using stratified sampling. To address class imbalance, class-weighted binary cross-entropy loss is incorporated during model training.

Four deep learning architectures, namely Deep Neural Network (DNN), one-dimensional Convolutional Neural Network (1D-CNN), Long Short-Term Memory (LSTM), and a Hybrid CNN-LSTM model, are developed and evaluated. The models are compared using Accuracy, Precision, Recall, F1-Score, and ROC-AUC to analyze their effectiveness in identifying potential churn customers. Among the evaluated models, the Hybrid CNN-LSTM achieved the highest test accuracy of 70.26%, while the DNN achieved an F1-Score of 57.38% and ROC-AUC of 75.85%.

To improve prediction interpretability, the framework incorporates explainable AI techniques based on Gradient × Input feature attribution to identify the factors contributing to churn predictions. The system also converts predicted churn probabilities into Low, Medium, and High risk categories, enabling customer prioritization. An interactive Streamlit dashboard is developed to present EDA results, model comparisons, individual predictions, batch predictions, risk levels, and feature importance in an accessible format.

The proposed framework integrates deep learning, customer behavior analysis, risk scoring, explainable AI, and interactive visualization into a unified churn prediction system. The resulting system provides

both predictive insights and interpretable information that can assist organizations in identifying high-risk customers and supporting data-driven customer retention strategies.

Signature of the Guide HOD